A First Course in Abstract Algebra

Section 24: Noncommutative Examples

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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24 Noncommutative Examples

Exercise 24.1: Addition in a three-element group algebra

Add the coefficients of each basis element, reducing in $\mathbb{Z}_5$:

$$(2e + 3a + 0b) + (4e + 2a + 3b) = 6e + 5a + 3b = \boxed{e + 0a + 3b}.$$

The group elements are formal basis labels; coefficients are combined only when their labels agree.

Exercise 24.2: Multiplication in a three-element group algebra

Since G is cyclic of order 3, its multiplication satisfies $a^2 = b$, $ab = ba = e$, and $b^2 = a$. Distribute in the given order:

$$(2e + 3a)(4e + 2a + 3b) = 8e + 4a + 6b + 12a + 6b + 9e.$$

Collecting coefficients gives $17e + 16a + 12b$. Reduction modulo 5 yields $\boxed{2e + a + 2b}$.

Exercise 24.3: A fourth power in the group algebra

Put $S = e + a + b$. For each fixed product label, exactly three of the nine terms in S^2 contribute to it, so $S^2 = 3S$. It follows that $S^4 = 9S^2 = 27S$. Therefore

$$(3S)^4 = 3^4 \cdot 27S = 2187S = 2S \quad \text{over } \mathbb{Z}_5.$$

The answer is $\boxed{2e + 2a + 2b}$.

Exercise 24.4: An ordered quaternion product

Use $i^2 = j^2 = k^2 = -1$, $ij = k$, $jk = i$, $ki = j$, and reverse these last three products with a minus sign. Then

$$(i + 3j)(4 + 2j - k) = 4i + 2k + j + 12j - 6 - 3i.$$

Collecting the four real coordinates gives $\boxed{-6 + i + 13j + 2k}$. Keeping the order of factors avoids the wrong signs in $ik = -j$ and $jk = i$.

Exercise 24.5: Quaternion powers and multiplication

Reduce the powers first: $i^2 = -1$, $j^3 = -j$, and $i^5 = i$. Thus

$$i^2 j^3 k j i^5 = (-1)(-j)kji = (jk)ji = iji = ki = \boxed{j}.$$

Each reassociation uses associativity, and no two noncommuting factors have been interchanged.

Exercise 24.6: The inverse of a pure quaternion

For $q = i + j$, its conjugate is $\overline{q} = -i - j$ and its squared norm is $1^2 + 1^2 = 2$. Hence

$$\boxed{(i+j)^{-1} = \frac{-i-j}{2}}.$$

For a direct check, $(i+j)^2 = i^2 + ij + ji + j^2 = -2$, so multiplication by the displayed inverse gives 1 on either side.

Exercise 24.7: The inverse of a quaternion product

Multiply the factors before inverting:

$$(1+3i)(4j+3k) = 4j + 3k + 12k + 9ik = -5j + 15k.$$

The squared norm is $25 + 225 = 250$, and the conjugate is $5j - 15k$. Therefore

$$\boxed{[(1+3i)(4j+3k)]^{-1} = \frac{j-3k}{50}}.$$

Indeed a pure quaternion $v = -5j + 15k$ has $v^2 = -250$, so $v(-v/250) = 1 = (-v/250)v$.

Exercise 24.8: A product in the group algebra of the symmetric group

Discard the zero coefficients. The factors are $A = \rho_1 + \mu_2 + \mu_3$ and $B = \rho_0 + \rho_1 + \mu_1 + \mu_3$. Using the source's right-to-left permutation composition, the required twelve products are

	ρ_0	ρ_1	μ_1	μ_3
ρ_1	ρ_1	ρ_2	μ_3	μ_2
μ_2	μ_2	μ_3	ρ_2	ρ_1
μ_3	μ_3	μ_1	ρ_1	ρ_0

.

The multiplicities of $\rho_0, \rho_1, \rho_2, \mu_1, \mu_2, \mu_3$ are $1, 3, 2, 1, 2, 3$. Coefficients are in $\mathbb{Z}_2$, so even multiplicities disappear and odd ones become 1. Thus

$$\boxed{AB = \rho_0 + \rho_1 + 0\rho_2 + \mu_1 + 0\mu_2 + \mu_3}.$$

Exercise 24.9: The center of the nonzero quaternion group

Every nonzero real scalar commutes with every quaternion, so $\mathbb{R} \setminus \{0\}$ is contained in the center. Conversely let $q = a + bi + cj + dk$ be central. Comparing

$$qi = -b + ai + dj - ck, \qquad iq = -b + ai - dj + ck$$

gives $c = d = 0$. Then comparing $(a+bi)j = aj + bk$ and $j(a+bi) = aj - bk$ gives $b = 0$. Therefore $q = a$ is real and nonzero. The center is exactly $\boxed{\mathbb{R} \setminus \{0\}}$ under multiplication.

Exercise 24.10: Two other complex planes inside the quaternions

Take $F_j = \{a + bj : a, b \in \mathbb{R}\}$ and $F_k = \{a + bk : a, b \in \mathbb{R}\}$. Because $j^2 = k^2 = -1$, their induced addition and multiplication obey exactly the complex-number formulas. The maps $a + bi \mapsto a + bj$ and $a + bi \mapsto a + bk$ are bijective ring homomorphisms from $\mathbb{C}$, hence field isomorphisms. These two subsets differ from each other and from the usual $\{a + bi\}$: for example j belongs only to F_j among these three planes.

Exercise 24.11: True or false: noncommutative examples

(a) False: $E_{11}E_{22} = 0$ in $M_2(F)$, with both factors nonzero.

(b) False: $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ is nonzero but has no inverse.

(c) False as stated for every abelian group: for the trivial group, the identity endomorphism equals the zero endomorphism. For every nontrivial A, the identity map is a nonzero unity.

(d) False: $A = \mathbb{Z}$ has a nonzero identity endomorphism as unity.

(e) False: for nontrivial A, the zero endomorphism is not an automorphism, so these automorphisms do not even contain the additive identity of a subring.

(f) False: take $R = \mathbb{Z}_2$. The group ring of the infinite cyclic group then has characteristic 2, whereas $\mathbb{Z}$ has characteristic zero, so the rings are not isomorphic.

(g) True: both the scalar coefficients and group labels commute, so every pair of formal sums commutes on expansion.

(h) False: $ij = k$ but $ji = -k$, so multiplication is not commutative. They form a division ring.

(i) True: quaternion multiplication is associative, 1 is the identity, and $q^{-1} = \overline{q}/\|q\|^2$ exists for every nonzero q.

(j) False: the real scalars, and also the usual complex plane, are subrings that are fields.

Exercise 24.12: Two failures of commutative-field conclusions

(a) In $\mathbb{H}$, the degree-two polynomial $x^2 + 1$ vanishes at $i, -i, j, -j, k, -k$, already six distinct zeros. More generally every $bi + cj + dk$ with $b^2 + c^2 + d^2 = 1$ has square -1. Thus the commutative-field root bound fails in a strictly skew field.

(b) The eight quaternions $\{\pm 1, \pm i, \pm j, \pm k\}$ form a finite multiplicative subgroup. Every element has order at most 4, so none generates all eight elements. This quaternion group is not cyclic.

Exercise 24.13: The other zero-divisor equation for an endomorphism

The source map is $\phi(m, n) = (m + n, 0)$. Define $\theta(m, n) = (m, -m)$; this is a nonzero additive endomorphism of $\mathbb{Z} \times \mathbb{Z}$. Then

$$(\phi\theta)(m, n) = \phi(m, -m) = (0, 0)$$

for every (m, n). Thus $\phi\theta = 0$ with $\theta \neq 0$, giving the required zero product on the side opposite to the source example's $\psi\phi = 0$.

Exercise 24.14: Six units over any field

The six matrices

$$\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}, \quad \begin{pmatrix}1 & 1\\0 & 1\end{pmatrix}, \quad \begin{pmatrix}0 & 1\\1 & 0\end{pmatrix}, \quad \begin{pmatrix}0 & 1\\1 & 1\end{pmatrix}, \quad \begin{pmatrix}1 & 1\\1 & 0\end{pmatrix}, \quad \begin{pmatrix}1 & 0\\1 & 1\end{pmatrix}$$

are distinct because $0 \neq 1$ in a field. Their determinants are $1, 1, -1, -1, -1, 1$, respectively, all nonzero even in characteristic 2 (where $-1 = 1$). The invertible-matrix criterion proves that all six are units of $M_2(F)$.

Exercise 24.15: Endomorphisms of cyclic additive groups

An additive endomorphism of $\mathbb{Z}$ is uniquely determined by the integer $k = \phi(1)$, and then $\phi(n) = kn$ for every integer n. The map $k \mapsto \phi_k$ is a bijection $\mathbb{Z} \to \mathrm{End}(\mathbb{Z})$, with

$$\phi_k + \phi_l = \phi_{k+l}, \qquad \phi_k \circ \phi_l = \phi_{kl}.$$

Thus it is a ring isomorphism.

For $\mathbb{Z}_n$, the same argument uses $k = \phi([1]) \in \mathbb{Z}_n$ and $\phi([r]) = [r]k$. Every k defines a well-defined endomorphism, since $nk = 0$ automatically. Addition and composition again correspond to $k + l$ and kl modulo n, proving $\mathrm{End}(\mathbb{Z}_n) \cong \mathbb{Z}_n$.

Exercise 24.16: Endomorphisms of a two-dimensional binary group

View $\mathbb{Z}_2 \times \mathbb{Z}_2$ as a two-dimensional vector space over $\mathbb{Z}_2$. Every additive endomorphism is linear and is determined by arbitrary images of its two basis vectors. Each image has four choices, so there are $4^2 = 16$ endomorphisms. The componentwise product ring $\mathbb{Z}_2 \times \mathbb{Z}_2$ has only four elements. Different cardinalities rule out a ring isomorphism. Equivalently the endomorphism ring is $M_2(\mathbb{Z}_2)$, which is also noncommutative.

Exercise 24.17: The differentiation–multiplication commutator

Here $Xf = xf$ and $Yf = f'$ are endomorphisms of the additive polynomial group. For every polynomial f,

$$(YX - XY)f = (xf)' - xf' = f + xf' - xf' = f.$$

The product rule can be checked on each monomial x^n: its two terms are $(n+1)x^n$ and nx^n. Thus $YX - XY$ acts as the identity on every polynomial and equals the unity endomorphism $\boxed{1}$.

Exercise 24.18: The group ring of a one-element group

If $G = \{e\}$, every formal sum has the form re. The map $re \mapsto r$ is bijective. Furthermore $(re) + (se) = (r + s)e$ and $(re)(se) = (rs)e$, since $e^2 = e$. The map preserves both operations, proving $RG \cong R$.

Exercise 24.19: A matrix realization of the quaternions

Put

$$E = I_2, \qquad I = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad J = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

The i in these matrix entries is the complex imaginary unit.

(a) Since quaternion $ij = k$, multiplicativity forces

$$\boxed{K = IJ = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}}.$$

(b) With $IJ = K$ used as this definition, the other eight multiplication-table equations to check are

$$I^2 = J^2 = K^2 = -E, \qquad JI = -K, \qquad JK = I, \qquad KJ = -I, \qquad KI = J, \qquad IK = -J.$$

Direct matrix multiplication verifies them: the three squares are $\mathrm{diag}(-1,-1)$, $JI = \mathrm{diag}(-i,i)$, $JK = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, and the remaining reversed products have the stated signs. Identities involving E hold automatically. Real-linearity of the definition makes additivity automatic, and distributivity reduces multiplicativity for arbitrary quaternions to these basis products.

(c) We must also check injectivity, since surjectivity onto the image is automatic. The explicit matrix is

$$\phi(a + bi + cj + dk) = \begin{pmatrix} a + di & b + ci \\ -b + ci & a - di \end{pmatrix}.$$

If it is zero, real and imaginary parts of its entries give $a = d = b = c = 0$. Thus the kernel is zero, proving the required isomorphism onto $\phi[\mathbb{H}]$.